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The Relevance of Emerging Solutions for Thinking, Decision Making and Acting. The case of Smart Grids

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ABSTRACT

The paper presents a real novelty in Mathematical Modelling, which may have an enormous relevance in our way of Thinking, Decision Making and Acting. At the same time it would like to represent an explicit "Tribute" to Prof. Odum, because the original concept is already seminally present in his well-known Rules of Emergy Algebra.

This novelty in Mathematical Modelling is represented by the so-called "Emerging Solutions", which are radically different from solutions to traditional mathematical problems. This is because any traditional solution to an algebraic or differential problem is always represented by a formal expression that, when reintroduced into the initial formulation of the problem, reduces the latter to a perfect identity.

Emerging Solutions, on the contrary, show an Ordinal Information content which is always much higher than the corresponding content pertaining to the initial formulation of the problem. Emerging Solutions, in fact, originate from any physical problem when this is modelled in accordance with the Maximum Ordinality Principle, and thus understood in Ordinal Terms.

Such a property, which represents one of the most interesting aspects of the Maximum Ordinality Principle, then suggests we adopt a Generative way of Thinking when designing a new practical application. The same happens at the level of Will, that is at the level of Decision Making. Clearly, if we really want to take advantage of those "Emerging Exits" which arise from the physical behaviour of the system. Finally, at the level of Acting, if we are really interested in favouring any "emerging behaviour" of the system which is decisively capable to improve our design. For instance, to get the maximum intrinsic stability of the system, so as to prevent any possible disturbance that might significantly alter its expected behaviour.

All these aspects will be illustrated through the case of Smart Grids, with particular reference to their large scale "intrinsic" instability and their recognized strong vulnerability to "cyber" attacks.

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1. Introduction

The paper is substantially aimed at presenting a real novelty in Mathematics (understood as a "Formal Language"), that is the "Emerging Solutions". These represent one of the most interesting aspects of the Maximum Ordinality Principle which, on the other hand, is nothing but the reformulation of the Maximum Em-Power Principle once "deprived" of any reference to Classical Thermodynamics (such as Energy, Exergy, and so on). In this sense, the paper would also like to represent an explicit "Tribute" to Prof. Odum, because the concept of "Emerging Solutions" is already seminally present in his well-known Rules of Emergy Algebra (Brown and Herendeen, 1996).

The second part of the paper will present some Ostensive Examples of "Emerging Solutions" (thus referred to Mathematical Models) together with their correspondence to some "Emerging Phenomena", which can be understood as "Emerging Exits" from Generative Processes.

All these examples will then converge to show, in the third part of the paper, analogous aspects in the study case of Smart Grids.

Finally, the comparison between "Emerging Solutions" and "Emerging Exits" from Generative Processes will bring out the potential relevance of the former, in terms of Thinking, Decision Making and Acting, when designing a new practical application.

More precisely, the case of Smart Grids will show how the concept of "Emerging Solutions" is able to suggest the best strategies to optimize the System. In fact, because of their capability of describing foreseeable behaviours of the System, they "reveal" how to improve the well-known large scale "intrinsic" instability of Smart Grids and their recognized strong vulnerability to "cyber" attacks.

2. The concept of Emerging Solutions

The simplest way of presenting "Emerging Solutions" is that of making a comparison with traditional solutions, as synoptically shown in Table 1. Nonetheless, from a conceptual point of view,

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Table 1
Traditional Solutions vs "Emerging Solutions".

Traditional Solutions (TS)	"Emerging Solutions" (ES)
(1) TS are those solutions which originate from any traditional algebraic or differential problem	(1′) ES are those Solutions which emerge from any Ordinal Differential Problem formulated in terms of "incipient" derivatives
(2) Consequently, they are solutions to all those mathematical models which describe each given system in terms of its pertaining traditional physical Laws	(2′) More specifically, they are those Solutions which describe any given System according to an Ordinal Model which is always based on the Maximum Ordinality Principle (M.O.P.)
(3) TS are always represented by a formal expression that, when reintroduced into the initial formulation of the problem, reduces the latter to a perfect identity	(3′) ES progressively acquire their increasing Ordinality during the same solution process, so that, if reintroduced into the initial formulation of the Problem, the latter *does not reduce* to a perfect identity
(4) They are "solution" to a problem in the sense of "loosing a *knot*"	(4′) They are "solution" to an Ordinal Problem in the sense of "disclosing *a seed*"

Emerging Solutions can be termed as such precisely because *they always show an Ordinal Information content which is much higher than the corresponding content of the initial formulation of the Problem*.

As an introductory Ostensive Example it is worth mentioning the case of the "Three-Body Problem". In fact, if the formal Solution to this problem, obtained in terms of the M.O.P., is re-introduced into the initial formulation of the same, we get the formulation of a new Ordinal Problem, corresponding to a "Six-Body Problem" (and so on). This is because the "Inter-action" between the higher Ordinal Information content of the Solution and the lower information content of the initial Problem give rise to a sort of a Feed-Back of Ordinal Nature, which can evidently be seen as a generalized version of the well-known Feed-Back in Emergy Algebra. This is precisely because the concept of "Emerging Solutions" traces back to the same Rules of Emergy Algebra (Brown and Herendeen, 1996).

If we consider in fact the Rules of Emergy Algebra pertaining to the three fundamental processes (Co-production, Inter-action, Feed-back) schematically represented in Fig. 1, we can easily recognize that the *non-conservative* Algebra adopted substantially asserts that: (i) "1 + 1 = 2 + something else" (in a Co-production); (ii) whereas "1 times 1 = 1 + something extra", where this "extra" strictly depends on the nature of the process (Inter-action or Feed-back, respectively).

In this sense Transformity (see Appendix A) may (also) be interpreted as a "cipher" of the internal self-organizing "activity" of the System (where the term "cipher" is here understood in a gnosiological sense). It would thus indicate that: *there are processes, in Nature, which cannot be considered as being pure "mechanisms"*.

In other words they are not describable in mere functional terms, because their outputs show an unexpected "excess" with respect to their pertinent inputs. Such an "excess" can be termed as *Quality* (with a capital Q) exactly because it is no longer understood as a simple "property" or a "characteristic" of a given phenomenon, but as being any *emerging* "property" (from the considered Process) *never reducible* to its phenomenological premises or to our traditional mental categories (Giannantoni, 2009, 2010a; see also Anderson, 1972).

This is precisely the sense of Transformity when this is understood as a "cipher" of the internal self-organizing "activity" of the System. This is also the reason why, according to the convention adopted for the term *Quality*, all the fundamental terms referable to such a concept are usually *capitalized* in this paper, so as to systematically remind their fundamental relationship to that concept.

It is then evident that, when transforming such a non-conservative Algebra (which is valid in steady-state conditions) to *dynamic conditions*, we end up by introducing a corresponding *non-conservative* Differential Calculus (the transpositions of the Rules in terms of "incipient" derivatives are shown by expressions (1.1), (1.2), and (1.3) in Fig. 1). This is precisely because the traditional derivative (as clearly illustrated in Appendix A) is not properly apt to represent such a concept of "cipher".

3. Emerging Solutions from the Maximum Ordinality Principle

The Maximum Ordinality Principle (Giannantoni, 2010a) is nothing but the reformulation of the Maximum Em-Power Principle (Odum, 1994a,b,c) given in a more general form by means of a new concept of derivative, the "incipient" derivative, whose mathematical definition has already been presented in Giannantoni (2001a,b, 2002, 2004, 2008a, 2009, 2010a) and Giannantoni and Zoli (2009a) and it is here synthetically recalled in Appendix A. In this way both Emergy and Transformity are replaced by the concept of Ordinality. This is why the principle was renamed as the Maximum Ordinality Principle (Giannantoni, 2010a,b). Its corresponding enunciation then becomes: "Every System tends to Maximize its own Ordinality, including that of the surrounding habitat". In formal terms

$$\left(\frac{\tilde{d}}{\tilde{d}t}\right)^{(\tilde{m}/\tilde{n})} \{\tilde{r}\}_s = 0 \quad \left(\frac{\tilde{m}}{\tilde{n}}\right) \rightarrow Max \tag{2}$$

where $(\tilde{d}/\tilde{d}t)$ is the symbol of the incipient derivative; $(\tilde{m}/\tilde{n})$ is the Ordinality of the System, which represents the Structural Organization of the same in terms of Co-Productions, Inter-Actions, Feed-Backs, while $\{\tilde{r}\}_s$ is the proper Space of the System (see Eq. (11) and related paragraph).

Eq. (2), considered together with its associated initial conditions, leads to the following explicit Solution

$$\{\tilde{r}\}_s = e \left\{ \begin{matrix} \tilde{\alpha}_{11}(t) & \tilde{\alpha}_{12}(t) & \dots & \tilde{\alpha}_{1n}(t) \\ \tilde{\alpha}_{21}(t) & \tilde{\alpha}_{22}(t) & \dots & \tilde{\alpha}_{2n}(t) \\ \dots & \dots & \dots & \dots \\ \tilde{\alpha}_{m1}(t) & \tilde{\alpha}_{m2}(t) & \dots & \tilde{\alpha}+_{mn}(t) \end{matrix} \right\}. \tag{3}$$

Strictly speaking, enunciation (2) formally asserts that: "every System is a Self-organizing System which persistently propends towards the Maximum Ordinality conditions". However, when a Self-organizing System *effectively reaches* such very special conditions, it presents itself as being self-structured in a radically different way with respect to its initial Ordinality. This is because the latter has evolved according to the following Trans-formation

$$\left(\frac{\tilde{m}}{\tilde{n}}\right) \rightarrow \{\{\tilde{2}/\tilde{2}\} \uparrow \{\tilde{2} \uparrow\}\} \uparrow \tilde{N}, \tag{4}$$

where $\{\tilde{2}/\tilde{2}\}$ represents a "binary-duet" coupling; the Ordinal power $\{\tilde{2} \uparrow\}$ indicates the "perfect specularity" of the previous "binary-duet" structure, while $\uparrow \tilde{N}$ indicates the Ordinal Over-structure of the $\tilde{N}$ elements of the System considered as a Whole (this is the reason for the "tilde" notation) (see also Giannantoni, 2009, 2010a,b). More precisely, Trans-formation (4) is due to the correlative circumstance that, under Maximum Ordinality conditions, the System also achieves its *Maximum Internal Stability*.

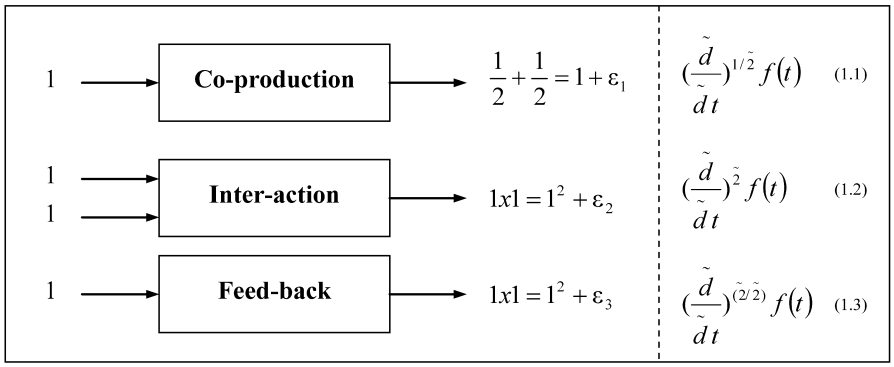

Fig. 1. Transformity as a "cipher" of the internal self-organizing activity of the System and corresponding representation under Dynamic Conditions in terms of "incipient" derivatives.

This corresponds to the fact that *each single couple* of elements structures itself in a "binary-duet" relationship, according to the evolution described by the following equation

$$\left(\frac{\tilde{d}}{\tilde{d}t}\right)^{(\tilde{2}/\tilde{2})}\left\{\{\tilde{r}\}\otimes\left(\frac{\tilde{d}}{\tilde{d}t}\right)^{(\tilde{2}/\tilde{2})}\{\tilde{r}\}\right\}=0. \tag{5}$$

Consequently, while the genesis of the Ordinal Structure of the System (expressed by solution (3)) is only due to Eq. (2), the generating Process of the internal Ordinal Stability of the System is expressed by Eq. (5), which, for the sake of simplicity and clarity, has been formulated with reference to any *single couple* of elements that progressively structure in a "binary-duet" Relationship.

Eq. (5) formally asserts that the proper Space of the System (now considered as being made up of two sole elements) is coupled with its specific Generativity in such a way as to originate a comprehensive Generative Capacity which is always in equilibrium.[1] Such an equation is precisely that which leads to the afore-mentioned *perfect specularity* which, in the case of two sole elements, is represented by the Ordinal structure $\{\{\tilde{2}/\tilde{2}\}\uparrow\{\tilde{2}\uparrow\}\}$, while in the case of $\tilde{N}$ elements is represented by the right hand side of Eq. (4).

Under these conditions the solution to Eq. (2) (and associated Eq. (5)) can be expressed in the form of a new exponential Ordinal Matrix

$$\{\tilde{r}\}_s=e\begin{Bmatrix}0 & \tilde{\alpha}_{12}(t) & \dots & \tilde{\alpha}_{1N}(t)\\ \tilde{\alpha}_{21}(t) & 0 & \dots & \tilde{\alpha}_{2N}(t)\\ \dots & \dots & \dots & \dots\\ \tilde{\alpha}_{N1}(t) & \tilde{\alpha}_{N2}(t) & \dots & 0\end{Bmatrix}, \tag{6}$$

where all the elements of the main diagonal are equal to zero, whereas all the other elements $\tilde{\alpha}_{ij}(t)$ satisfy the following *specularity* Relationships

$$\{\tilde{\alpha}_{ij}(t)\}^{\{\{\tilde{2}/\tilde{2}\}\uparrow\{\tilde{2}\uparrow\}\}}\overset{*}{=}\{\tilde{\alpha}_{ji}(t)\}^{\{\{\tilde{2}/\tilde{2}\}\uparrow\{\tilde{2}\uparrow\}\}}, \tag{7}$$

which represent a much more profound concept than the traditional symmetry (correspondingly, the symbol "$\overset{*}{=}$" indicates a simple assignation condition).

Moreover, the Generative Process that leads the System to its Maximum Ordinality and, at the same time, to its Maximum Stability conditions, also restructures the internal relationships between the various elements in such way as these show an additional "*emerging*" *property*, expressed by the following Harmony Relationships:

$$\tilde{\lambda}_{12}\oplus\tilde{\alpha}_{12}(t)\overset{*}{=}\tilde{\lambda}_{1j}\oplus\tilde{\alpha}_{1j}(t)\quad\text{for}\quad j=3,4,\dots,N \tag{8}$$

together with all their associated *incipient* derivatives, up to the order $N-1$

$$\{\overset{\circ}{\tilde{\lambda}}_{12}\oplus\overset{\circ}{\tilde{\alpha}}_{12}(t)\}^{\tilde{k}}\overset{*}{=}\{\overset{\circ}{\tilde{\lambda}}_{1j}\oplus\overset{\circ}{\tilde{\alpha}}_{1j}(t)\}^{\tilde{k}}\quad\text{for}\quad k=1,2,\dots,N-1, \tag{9}$$

where $\tilde{\lambda}_{ij}$ represent their corresponding *internal reciprocal* Correlating Factors.

This means that all the elements of Ordinal Matrix (6) can be obtained on the basis of 1 sole couple $\tilde{\alpha}_{ij}(t)$ and $N-1$ associated Correlating Factors.

As a consequence of all these successive formal passages the Solution to the System, originally structured in form (3), ends up by *progressively* assuming the following more general "emerging" form

$$\{\tilde{r}\}_s=e\begin{Bmatrix}0 & \tilde{\lambda}_{12}(t)\oplus\tilde{\alpha}_{12}(t) & \dots & \tilde{\lambda}_{12}(t)\oplus\tilde{\alpha}_{1N}(t)\\ \tilde{\lambda}_{21}(t)\oplus\tilde{\alpha}_{21}(t) & 0 & \dots & \tilde{\lambda}_{2N}(t)\oplus\tilde{\alpha}_{2N}(t)\\ \dots & \dots & \dots & \dots\\ \tilde{\lambda}_{N1}(t)\oplus\tilde{\alpha}_{N1}(t) & \tilde{\lambda}_{N2}(t)\oplus\tilde{\alpha}_{N2}(t) & \dots & 0\end{Bmatrix}. \tag{10}$$

In this respect, it is worth pointing out more clearly such a sequence of progressively "Emerging Solutions": (i) from initial form (3), which characterizes the System in its natural tendency towards the Maximum Ordinality Conditions; (ii) we pass to Solution (6) which expresses that, when the System effectively reaches such special conditions, it has also achieved its Maximum Stability conditions (Eq. (7)), through a Generating Process described by Eq. (5); (iii) the latter Process, however, does not characterize the sole single couples $\tilde{\alpha}_{ij}(t)$ (see Eq. (8)), but also all their associated "incipient" derivatives (see Eq. (9)); (iv) the latter are exactly those Ordinal Relationships that lead to final Solution (10), which shows that the System is also characterized by an additional "emerging" property: its internal *Ordinal Harmony*.

4. From Emerging Solutions to Emerging Exits

While "Emerging Solutions" are understood as a *mathematical* concept, the expression "Emerging Exits" specifically refers to those *phenomenological* aspects (or properties) which appear as being *not reducible* to traditional physical Laws.

In this section we will recall three examples of Emerging Solutions that can be obtained by adopting the M.O.P. as a reference modelling criterion. Correspondingly, we will also bring out the

[1] The symbol $\otimes$ represents a more general form of the "vector" product. However, in this specific context, it can be assumed as being perfectly equivalent to the traditional vector product.

associated phenomenological "Emerging Exits", which can easily be recognized and interpreted on the basis of the former.

The first Ostensive Example (see Table 2) is represented by the well-known "Three-Body Problem", explicitly dealt with in (Giannantoni, 2007, ch. 5) and also recalled in (Giannantoni, 2008a, 2010a,b).

As is well-known, such a problem does not admit an explicit solution, not even in a closed form (as proved by Poincaré in 1888). Vice versa, when modelled on the basis of the M.O.P., it presents an explicit solution which can easily be extended to N bodies (Giannantoni, 2010b, 2011a,b). Such a solution presents itself as an "Emerging Solution" because it shows (among other aspects) the inapplicability of superimposition of spaces (see Table 2). Such a mathematical property corresponds, from a phenomenological point of view, not only to the recognized expansion of the Universe but, above all, to non-conservation of Energy (Giannantoni, 2010a). A result already anticipated by the same Poincaré, when asserted that "The conservation of Energy is a limitation imposed to *freedom* of complex systems" (Poincaré, 1952, p. 133), and also implicitly recognized by modern Astronomy through the "hypothesis" of "Dark Energy".

The second Ostensive Example is represented by Protein Folding, usually considered as being one of the most important "intractable" problems. In fact, although the problem is thought of as being theoretically solvable in principle, the time required in practice to be solved may range from hundreds to some thousands of years, even when run on the most updated computers (1 Petaflop). On the contrary, when modelled in adherence to the M.O.P., the Problem becomes "tractable", because it presents an explicit solution (Giannantoni, 2010a,b, 2011a). This, in turn, leads to an extremely significant reduction of the computation time (from thousand of years to some minutes), even by adopting a common PC, usually characterized by a much lower computation power (about 1 Gigaflop).

In such a case the explicit solution presents itself as an "Emerging Solution" because, among other aspects, is able to give the Harmonic spatial configuration of the folded Protein, whose topology is described by the Ordinal roots of Unity (see Eq. (25)). What's more, this "Emerging Solution" also paves the way to recognize that the well-known *mono-chirality* of Proteins is an "Emerging Exit" of their Generative Folding Process (Giannantoni, 2008a, ch. 18, 2010b). A phenomenological property which, as is well-known, still remains unexplained from its first discovery (in the early 1930s), precisely because it results as being *not reducible* to traditional categories and consequential theoretical approaches.

The Third Ostensive Example is represented by the well-known "Three Good Two Factor Problem", a famous problem of Neo-Classical Economics which does not admit an explicit solution, not even in a closed form. Vice versa, when modelled in adherence to the M.O.P. also this Problem presents an explicit Solution, even when it is extended to the case of "N-Good Three Factor Problem" (Giannantoni, 2011b). The Solution so obtained can still be considered as an "Emerging Solution" because it shows that, from a mathematical point of view, Maximum Economic Ordinality represents something "more" than the traditional concept of Pareto's optimality. Such an "Emerging Solution" then enables us to recognize, as corresponding "Emerging Exits", the existence of Ordinal Benefits which, on the contrary, are never accounted for by traditional GDP. This happens not only in steady state conditions, as already shown on the basis of the same Maximum Em-Power Principle (Giannantoni, 2009; Giannantoni and Zoli, 2010b), but also under dynamic conditions, on the basis of the Maximum Ordinality Principle (Giannantoni, 2011b).

This latter example is also particular apt to show the Relevance of Emerging Solutions for Thinking, Decision Making and Acting with specific reference to the case study of "Smart Grids".

To this purpose it is worth recalling some basic aspects pertaining to the possibility of a *direct transposition* of Emerging Solutions between two (or more) different spaces of analysis.

5. Transposition of models among proper spaces of analysis

The search for solution (10), *both in Classical and Quantum Mechanics* (think of "The N-Body Problem" and "Protein Folding", respectively) is facilitated not only by the structure of Eqs. (2) and (4), but also (and especially) by the conception of the basic reference space $\{\tilde{r}\}_s$, which is understood as *one sole entity*. This is why it can more appropriately be represented as follows

$$\{\tilde{r}\}_s = \{\tilde{x} \circledR \vec{i} \oplus \tilde{y} \circledR \vec{j} \oplus \tilde{z} \circledR \vec{k}\}, \tag{11}$$

where the coordinates $(\tilde{x}, \tilde{y}, \tilde{z})$ are understood as representing the exit of a Generative Process (this is the reason for the tilde notation) and the symbols $\oplus$ and $\circledR$ express more intimate relationships between the same: both in terms of sum ($\oplus$) and in terms of (relational) product ($\circledR$) (see later on) with respect to the traditional versors $\vec{i}, \vec{j}, \vec{k}$. However, for practical purposes, it is more useful to adopt the representation obtainable from a generalized version of Moivre's formula

$$\{\tilde{r}\}_s = \{\tilde{\rho} \circledR \tilde{i} \circledR e^{\tilde{\varphi} \circledR \tilde{j}} \circledR e^{\tilde{\vartheta} \circledR \tilde{k}}\}, \tag{12}$$

where the coordinates $\{\tilde{\rho}, \tilde{\varphi}, \tilde{\vartheta}\}$ are still considered as being representing the exit of a Generative Process, whereas the traditional $\vec{i}, \vec{j}, \vec{k}$ are now replaced by three unit spinors $\tilde{i}, \tilde{j}, \tilde{k}$, which are defined in such a way as to satisfy the following Relational Product Rules:

$$\tilde{i} \circledR \tilde{i} = \oplus 1 \qquad \tilde{i} \circledR \tilde{j} = \tilde{j} \qquad \tilde{i} \circledR \tilde{k} = \tilde{k} \tag{13.1}$$

$$\tilde{j} \circledR \tilde{i} = \tilde{j} \qquad \tilde{j} \circledR \tilde{j} = \Theta 1 \qquad \tilde{j} \circledR \tilde{k} = \tilde{k} \tag{13.2}$$

$$\tilde{k} \circledR \tilde{i} = \tilde{k} \qquad \tilde{k} \circledR \tilde{j} = \tilde{k} \qquad \tilde{k} \circledR \tilde{k} = \Theta 1. \tag{13.3}$$

Representation (12) is similar (albeit not strictly equivalent) to a system of three complex numbers, characterized by one real unit ($\tilde{i}$) and two imaginary units ($\tilde{j}$ and $\tilde{k}$).

5.1. Transposition of the "N Body Problem" to the "N Good Problem"

The proper Space of the System $\{\tilde{r}\}_s = \{\tilde{\rho}, \tilde{\varphi}, \tilde{\vartheta}\}$, which is typical of *both Classical and Quantum Mechanics*, can easily be transposed to *Economics* by simply adopting three new variables $\{\tilde{K}, \tilde{L}, \tilde{N}\}$, where $\tilde{K}$ = Capital, $\tilde{L}$ = Labour and $\tilde{N}$ stands for "Natural Resources". In this case, the adoption of "three" productive factors (instead of the traditional "two" factors usually considered in Neo-Classical Economics (NCE), namely Capital and Labour) is not only due to the fact that one of the major criticisms addressed to NCE is that of neglecting Nature as the *third fundamental factor* and, consequently, the *intrinsic value* of Natural Resources. It is especially due to the fact that *three* distinct variables enable us to represent, through their Ordinal Relationships, the three fundamental Processes pointed out by Prof. Odum (Co-Production, Inter-action, Feed-Back).

Under such conditions the transposition of the concepts to *Economics* clearly shows that any Good (i), represented in the Space of Goods as

$$\{\tilde{r}\}_{G,i} = \{\tilde{K}_i, \tilde{L}_i, \tilde{N}_i\}, \tag{14}$$

constitutes *one sole entity* and, at the same time, represents something "*extra*" with respect to the simple "sum" of its productive factors. This evidently reflects the *Holistic* Approach subjacent to the Maximum Ordinality Principle, which is ever-present in all the developments presented in this paper.

Table 2
Mathematical Models, corresponding Emerging Solutions and associated Emerging Exits.

Mathematical Models based on the M.O.P.	Emerging Solutions (to M.O.P. Mathematical Models)	Emerging Exits (phenomenological properties)
(1) The Three (N) Body Problem presents an explicit solution	Non-superimposition of Spaces $\{\tilde{r}\}_s \neq \sum \{\tilde{r}\}_i$	Expansion of the Universe $\{\tilde{E}n\}_s \neq const$
(2) Protein Folding becomes a tractable problem	Harmonic spatial configuration based on the Ordinal Roots of Unity	Mono-chirality of Proteins
(3) The Three (N) Good Problem presents an explicit solution	Maximum Ordinality is much more than Pareto's optimality	Ordinal Benefits (not accounted for by GDP)

On the basis of such a transposition, the "N Good Problem" can still be formulated in terms of the Maximum Ordinality Principle (see Eq. (2)), in order to obtain the explicit general solution (10), which is now understood in the corresponding proper Space of Goods $\{\tilde{r}\}_G = \{\tilde{K}, \tilde{L}, \tilde{N}\}$ (see (14)).

If we now consider N different goods, characterized by the arbitrary values $\{\tilde{K}_i, \tilde{L}_i, \tilde{N}_i\}$, there is no certainty that (at least in principle) their "coordinates" satisfy all the afore-mentioned Harmony conditions. Vice versa, when the production of N Goods is more appropriately considered as an "Emerging Exit" of Generative Processes, it represents a unique Ordinal entity, because the N Goods are "harmoniously" coordinated among themselves according to Relationships (8) and (9).

Under these conditions, in fact, any Ordinal set of Goods is not a simple arithmetical "sum" of the same, but gives origin to something "extra": a *unique* and *irreducible* entity.

Such a property appears as being decisively important for any Decision Maker, because it suggests the best strategy that is effectively able to maximize the Ordinality of the Economic System analyzed and, at the same time, to reduce the exploitation of Natural Resources.

In fact, if the considered Economic System is not at its Maximum Stability conditions, in the presence of an eventual *new* good produced, the comprehensive System (now made up of $N+1$ goods) might evolve towards a progressively lower level of Ordinality. This more or less markedly depends on the initial conditions of the new good added. In many cases, however, the System could even become manifestly *unstable*.

If, on the contrary, the System is at its Maximum Ordinal Stability, the new additional good may or may not respect the Harmonic Conditions of the comprehensive System as a Whole. In the former case the Economic System will always *evolve* towards a higher level of Ordinality (and associated new Maximum Stability). In the latter case the System will always remain *stable* (albeit at a slightly lower level), because of its capacity (as a Whole) of "assimilating" the external "disturbance", as a consequence of the coordinated generation of Goods due to *all* the harmonious Ordinal Inter-Relationships between the N pre-existing Generative Processes.

5.2. Transposition of the same concepts from Economics to Smart Grids

Let us now consider a Smart Grid, understood as a Production System made up of N electrical Generators, characterized by three parameters: V_j = voltage, I_j = maximum current intensity, Φ_j = internal impedance phase (see Fig. 2).

As a consequence of the Energy Conservation Principle we usually assume that

$$I_{tot}(t) = \sum_{j=1}^{N} (I_j \ \sin \omega t) = \left(\sum_{j=1}^{N} I_j \right) \cdot \sin \omega t. \tag{15}$$

In reality there is always a distortion "drift", because we are dealing with N *Generative* Processes. Consequently, if we "sum" their corresponding currents, now considered as "Emerging Exits" of the same Processes (thus characterized by the tilde notation), we always have that

$$\{\tilde{I}_{tot}(t)\} = \{\tilde{I}_1(t) \oplus \tilde{I}_2(t) \oplus \ldots \tilde{I}_n(t)\} \neq \left(\sum_{j=1}^{N} I_j \right) \cdot \sin \omega t. \tag{16}$$

Inequality (16) can easily be understood by considering the simple case of two generators (j and k), and the corresponding "sum" of their corresponding currents, which gives

$$\{\tilde{I}_j(t) \oplus \tilde{I}_k(t)\} \neq I_j(t) + I_k(t) = (I_{j,m} + I_{k,m}) \cdot \sin \ \omega t. \tag{17}$$

In fact Taylor's expansion series of the function on the right hand side of inequality (17), rewritten as

$$f(t_0 + \Delta t) = I_j(t) + I_k(t) = e^{\ln[I_j(t) + I_k(t)]} = e^{\alpha(t)}, \tag{18}$$

where $\alpha(t) = \ln\{I_j(t) + I_k(t)\}$, gives

$$f(t_0 + \Delta t) = e^{\alpha(t_0)} + e^{\alpha(t_0)} \cdot \dot{\alpha}(t_0) \cdot \frac{\Delta t}{1!} + \sum_{k=2}^{n} \left[\frac{d^k e^{\alpha(t)}}{dt^k} \right]_{t_0} \frac{\Delta t^k}{k!} + \cdots = (I_{j,m} + I_{k,m}) \cdot \sin \omega t, \tag{19}$$

which (as is well known) represents a perfect sinusoidal trend.

Vice versa, if we consider the left hand side of inequality (17), where the "sum" refers to the same two currents understood as the corresponding Exits of two Generative Processes, we have an Ordinal Relationship, which can analogously be rewritten as

$$\tilde{f}(t_0 + \Delta t) = \{\tilde{I}_j(t) \oplus \tilde{I}_k(t)\} = e^{\ln\{\tilde{I}_j(t) \oplus \tilde{I}_k(t)\}} = e^{\tilde{\alpha}(t)}, \tag{20}$$

where $\tilde{\alpha}(t) = \ln\{\tilde{I}_j(t) \oplus \tilde{I}_k(t)\}$. In this case, "incipient" Taylor's expansion series of (20), which is namely obtained in terms of incipient derivatives (Giannantoni, 2009, 2010a), gives

$$\tilde{f}(t_0 + \Delta t) = e^{\alpha(t_0)} + e^{\alpha(t_0)} \cdot \overset{\circ}{\alpha}(t_0) \cdot \frac{\Delta t}{1!} + e^{\alpha(t_0)} \sum_{k=2}^{n} [\overset{\circ}{\alpha}(t_0)]^k \frac{\Delta t^k}{k!} \neq (I_{j,m} + I_{k,m}) \cdot \sin \ \omega t, \tag{21}$$

which is substantially different from a sinusoidal trend as a consequence of the differences (that appear for $n \geq 2$) between the corresponding terms of the two expansion series, as shown by Eq. (22) (Giannantoni, 2009, 2010a):

$$\sum_{k=2}^{n} \left[\frac{d^k e^{\alpha(t)}}{dt^k} \right]_{t_0} \cdot \frac{\Delta t^k}{k!} \neq e^{\alpha(t_0)} \sum_{k=2}^{n} [\overset{\circ}{\alpha}(t_0)]^k \cdot \frac{\Delta t^k}{k!}. \tag{22}$$

Such a distortion "drift" (with respect to a perfect sinusoidal trend) tends to amplify even under normal exercise conditions, as

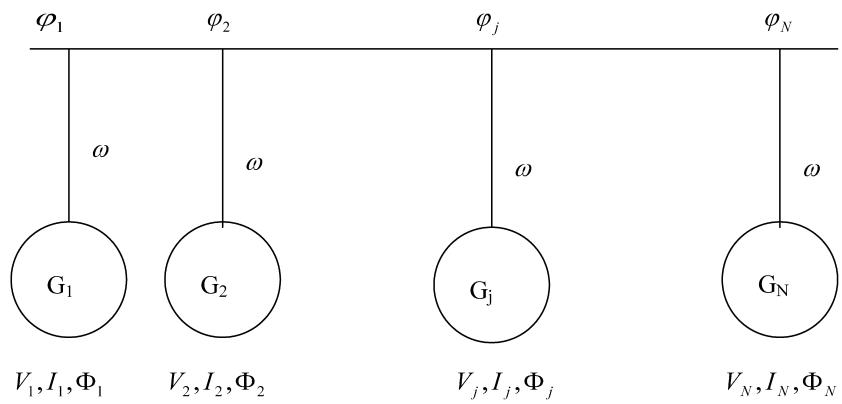

Fig. 2. A simplified scheme of a Production System made up of *N* electrical Generators.

a consequence of the different currents produced by the *N* generators, because of the differentiated time increases (or decreases) in the electrical charges to be supplied.

A "drift" which becomes even more marked in the case of a cyber attack. In fact, among the different forms of attacks (see Harris, 2010), we may simply think of the case in which such an external interference not only modifies the intensity of the maximum current of the single plant involved by the attack but, in a special way, tends to significantly change its phase (up to the opposite phase) (Harris, 2010).

On the contrary, if the Smart Grid is designed according to the M.O.P., the corresponding Maximum Stability conditions can always be assured. This is because the various plants are connected to each other in such a way to satisfy the afore-mentioned Harmony Relationships (8) and (9). The latter, under steady state conditions, assume the following explicit form (Giannantoni, 2011b, 2012, see also www.ordinality.org)

$$\{\tilde{V}_{1l}, \tilde{I}_{1l}, \tilde{\Phi}_{1l}\} \circledR (\sqrt[N-1]{\tilde{1}})_l \overset{*}{=} \{\tilde{V}_{12}, \tilde{I}_{12}\tilde{\Phi}_{12}\}, \tag{23}$$

where

$$\tilde{V}_{ji} = \tilde{V}_j - \tilde{V}_i, \qquad \tilde{I}_{ji} = \tilde{I}_j - \tilde{I}_i, \qquad \tilde{\Phi}_{ji} = \tilde{\Phi}_j - \tilde{\Phi}_i, \tag{24}$$

while ($\sqrt[N-1]{\tilde{1}}$) represents the $N-1$ Ordinal Roots of Ordinal Unity ($\tilde{1}$). That is:

$$(\sqrt[N-1]{\tilde{1}}) = \{\tilde{\alpha}_l \circledR \tilde{i} \oplus \tilde{\beta}_l \circledR \tilde{j} \oplus \tilde{\gamma}_l \circledR \tilde{k}\} \quad \text{for} \quad l = 1, 2, \ldots, N-1 \tag{25}$$

so that

$$\{\tilde{\alpha}_l \circledR \tilde{i} \oplus \tilde{\beta}_l \circledR \tilde{j} \oplus \tilde{\gamma}_l \circledR \tilde{k}\}^{N-1} = \tilde{1}. \tag{26}$$

Conditions (23) can evidently be generalized to any kind of exercise conditions. This means that the considered Smart Grid can always be controlled in such a way as to work at its Maximum Ordinal Stability, not only under normal exercise conditions, but also in the case of cyber attacks.

This is because the adoption of Eq. (23), understood as an "Emerging Solution" to the mathematical model of a Smart Grid based on the M.O.P., is able to foresee the corresponding phenomenological "Exits" of the physical System analyzed. This offers some important advantages:

(i) the possibility of optimizing the exercise conditions of any Smart Grids already realized;
(ii) the possibility of improving the design of any new Smart Grids to be realized;
(iii) in both cases, not only as far as the intrinsic stability of the Grid is concerned, but also (and especially) with reference to external disturbances (such as, for instance, cyber attacks).

6. Conclusions

The analysis previously presented points out that the traditional design of Smart Grids and their quantitative optimization (Energy produced, Energy supplied, associated costs, etc.) generally do not satisfy the criterion of their maximum intrinsic stability conditions, in a special way in the case of cyber attacks.

On the contrary, the analysis based on the Maximum Ordinality Principle (and its associated "Emerging Solutions") enables the Decision Maker to recognize in advance those (theoretical) optimal working conditions which realize the Maximum Ordinality of the System and, at the same time, to favour the corresponding "emerging behaviour" (as an "Emerging Exit"), which is decisively capable to improve the internal Ordinal Stability of the System with respect to any possible (internal or external) transient conditions.

In other terms, the introduction of the concept of "*Emerging Solutions*" suggests:

(i) we Think in Generative terms when designing any new practical application;
(ii) we Make Decisions in the respect of those solutions which are "emerging" from the mathematical model, in such a way as to take the maximum advantage from those corresponding "Emerging Exits" which are foreseen to be arising from the physical behaviour of the system;
(iii) we adopt consequential Actions for favouring that specific "emerging behaviour" of the System which appears as being decisively capable to improve our design.

In the particular case of Smart Grids, this means getting the Maximum Intrinsic Stability of the System, so as to prevent any possible disturbance that might significantly alter its expected behaviour.

Appendix A.

This short appendix is aimed at presenting: (i) the main analogies and differences between the Maximum Em-Power Principle (M.Em-P.P.) and its reformulation in terms of the Maximum Ordinality Principle (M.O.P.); (ii) the definition of the "incipient" derivative; (iii) an example of Emerging Solution; (iv) some advantages of the model proposed in Section 5.1 compared to other models based on Neo-classic Economics.

A.1. Analogies and differences between the Maximum Em-Power Principle and the Maximum Ordinality Principle

The Maximum Em-Power Principle	The Maximum Ordinality Principle
(i) The M.Em-P.P. assumes that there are three basic processes (Co-production, Inter-action, Feed-back), to which any other complex process can be reduced	(i′) The M.O.P. starts from the same assumption, but it also shows that, in complex systems, the three basic processes are always *co-present*, even in a subjacent form
(ii) The specificity of such fundamental processes can be represented by means of the particular structures of their different *outputs*	(ii′) The specificity of such basic processes, however, can be represented by means of their particular *internal generative structures* or, better, by their specific *self-organizing capacities*
(iii) To this purpose, Odum introduced the concept of Emergy, defined as "the total *solar* equivalent *available Energy* directly and indirectly *used up* to generate a specific form of Energy (or product)"	(iii′) To this purpose a *new concept of derivative* was introduced (termed as "incipient" derivative), characterized by a specific *Ordinality* (see later on)
(iv) In the definition of Emergy it is implicitly understood that its evaluation has to be done according to a particular *non-conservative* accountability technique represented by the Rules of Emergy Algebra (Brown and Herendeen, 1996)	(iv′) In this case, the *non-conservative* accountability technique, represented by the Rules of Emergy Algebra (generally valid in steady state conditions), is replaced by a new *non-conservative* Differential Calculus, which is applicable to any variable conditions
(v) The introduction of Emergy does not represent a violation of the Principles of Classical Thermodynamics. It simply shows something "new". That is, the fact that: any form of Energy "vehicles" something extra with respect to its mere *quantitative* aspects	(v′) The concept of Ordinality, which replaces the concept of Emergy (as well as that of Transformity) shows that is more appropriate to assert that: it is precisely the Ordinality of the System that which "guides" the corresponding associated cardinalities
(vi) Such an assertion is well-represented by the concept of Transformity, understood as a result of two *independent* Balances. The Transformity associated to each pathway (i) is in fact defined as (Brown and Herendeen, 1996) $Tr(y_i) = \frac{Em(y_i)}{Ex(y_i)}$	(vi′) In such a phenomenological description there are no more two independent Balances, but only one: that pertaining to the evolution of Ordinality, characterized by the adherent evolution of its corresponding cardinalities
where the denominator derives from Classical Thermodynamics, whereas the numerator is obtained by means of the pertaining Rules of the afore-mentioned *non-conservative* Emergy Algebra	This aspect is particular important because it is able to show (for instance) that, at a large scale (such as in the case of the Solar System, and even more in the case of the Universe), the assumption of Conservation of Energy represents a rough approximation

A.2. The "incipient" derivative

The mathematical definition of the "incipient" derivative (see Giannantoni, 2001a,b, 2002, 2004, 2008a, 2009, 2010a) is given by

$$\left(\frac{\tilde{d}}{\tilde{d}t}\right)^{Ord} f(t) = \widetilde{\lim_{\tilde{\Delta}t:0\to 0^+}} \circ \left(\widetilde{\frac{\tilde{\delta}-1}{\tilde{\Delta}t}}\right)^{Ord} \circ f(t), \tag{A1}$$

where *Ord* (which stands for Ordinality) is any rational number. Such a new derivative was intentionally termed as "incipient" precisely because it describes any given process in its generating activity or, equivalently, because it focuses on the pertinent outputs *in their specific act of being born*.

The structure of definition (A1) may appear as being substantially "similar" to the traditional derivative. In reality, it is deeply different. In this respect, the adoption of the "tilde" notation would in fact indicate that the same symbols are now understood in substantially different way.

Before illustrating definition (A1) in more details, it is worth recalling that the introduction of a new concept of "derivative" was required by the fact that the traditional derivative (d/dt) is nothing but the formal reflex of three fundamental pre-assumptions when describing physical-biological-social systems: (i) *efficient causality*; (ii) *necessary logic*; (iii) *functional relationships*. Such an aprioristic perspective thus excludes, from its basic foundation, the possibility that any process output might ever show anything "extra", with respect to its corresponding input, as a consequence of the intrinsic (supposedly) *necessary*, *efficient* and *functional* dynamics of the system analyzed. Consequently, such a theoretical approach will never see any "output excess", exactly because it has already excluded from the very beginning (but only aprioristically) that there might be "any". In this sense it is possible to say that such an approach describes all the phenomena as they were mere "mechanisms".

Co-production, Inter-action and Feed-back Processes, on the contrary, suggest we think of a different form of "causality", precisely because their outputs always show something in "excess" with respect to their inputs. This "causality" may be termed as "generative" causality or "spring" causality or whatsoever. In all cases the basic concept is rather clear. Any term adopted is simply finalized at indicating that it is worth supposing a form of "causality" which is capable of giving rise to something "extra" with respect to what it is usually foreseen (and expected) by the traditional approach.

The same happens for Logic. In fact, a different Logic is correspondently needed in order to contemplate the possibility of coming to conclusions much richer than their corresponding premises. This new form of Logic, in turn, could correspondently be termed as "adherent" Logic, because its conclusions are always faithfully conform to the premises. Nonetheless, the conclusions could even be well-beyond what is strictly foreseen by the same premises when interpreted in strictly necessary terms. More specifically, from a *formal* point of view, this exactly corresponds to the genesis of mathematical *Emerging Solutions* (see Section 2 of the text).

As an adherent consequence of both previous concepts, the relationships between phenomena cannot be reduced to mere "functional" relationships between the corresponding cardinal quantities. This is because such relationships always "manifest" something else, which leads us to term them as "Ordinal" relationships. The term "Ordinal" would thus explicitly remind us that each part of the System is related to the others exclusively because, above all, it is related to the Whole (or, even better, it is "ordered" to the Whole) by the *same*, *unique* and *irreducible* internal genetic process of the Whole. This is why the Whole represents something which is much more than the simple "sum" of their parts.

The new concept of derivative is thus nothing but the adherent "translation", in formal terms, of the three new gnoseological concepts: *Generative Causality*, *Adherent Logic*, *Ordinal Relationships*.

The proper meaning of definition (A1) can easier be illustrated by means of a direct comparison with the traditional derivative. To this purpose it is worth preliminarily noting that the traditional increment $\Delta f(t) = f(t+\Delta t) - f(t)$ can equivalently be expressed in terms of the "*operator*" δ, which represents the variation $\delta f(t) = f(t+\Delta t)$ of the analyzed property $f(t)$, that is

$$\Delta f(t) = f(t+\Delta t) - f(t) = (\delta - 1) f(t). \tag{A2}$$

Consequently, the ratio

$$\frac{\tilde{\delta}-1}{\tilde{\Delta}t} \tag{A3}$$

(in definition (A1)) substantially replaces the traditional incremental ratio $\Delta/\Delta t$. The symbol "tilde", however, should remind us that its meaning is now completely different.

In fact, the comparison between the "incipient" derivative and the traditional derivative immediately shows that the latter corresponds to an "operative" definition, because the priority of the *operators* that constitute its definition is understood *from right to left*, that is: (i) firstly the concept of function (which is assumed to be a primary concept); (ii) then the incremental ratio (of the supposedly known function); (iii) thirdly the *operation* of limit (referred to the result of the previous two steps).

The "incipient" derivative, on the contrary, is based on the *direct priority* of the order of the three elements that constitute its definition, that is, *from left to right*. This is why they acquire a completely different meaning.

Let us start from the symbol $\tilde{Lim}$. The etymological origin of the word can help us to understand the new concept: "Limit" comes from the Latin word "Limen", which means a "threshold". It could be a "threshold" of a door or of a "window", from which we observe and describe the considered phenomenon. In such a case the symbol $\tilde{\Delta}t : 0 \to 0^+$ indicates not only the initial time of our registration, but also the proper "*origin*" (in its etymological sense) of something new which we observe (and are going to describe) in *its proper act of being born*. It becomes then evident that the "operator" $\tilde{\delta}$ should be now understood as a "*generator*", that registers the variation of the observed property $f(t)$, not only in terms of quantity, but also, and especially, in terms of Quality (as the symbol "tilde" would expressly remind). Thus the ratio (A3) indicates not only a quantitative variation in time, but both the variation in Quality and quantity. In fact, from the very beginning of any Process we can recognize its specific genesis in the form of a Co-production, Inter-action, Feedback, respectively. We can then take explicit note of this genetic property by means of a rational number (*Ord*) as an exponent of the Ordinal Incremental Ratio: $1/\tilde{2}$, $\tilde{2}$, and $\{\tilde{2}/\tilde{2}\}$ respectively.

In other terms, the "incipient" derivative represents the *Generativity* (or, equivalently, the Generative Capacity) of the considered Process, phenomenologically represented by the output "excess" (per unit time) characterized by both its Ordinality and its related cardinality. This is also the reason why the sequence of the symbols (in Eq. (A1)) is interpreted as a generative inter-action (see the symbol "$\circ$") between the three considered concepts. In this way the "incipient" derivative is able to *unify* and, at the same time, to *specify*, the three basic Processes, now explicitly understood in terms of Quality.

The Generativity concept, in fact, is that which unifies the three Processes, whereas the pertinent Ordinality expresses the structure of the corresponding output Relationships (as "binary", "duet", and "binary-duet" Relationships, respectively), which are consequently understood as a Whole. This is because, when we take the incipient (or "prior") derivative of order $1/\tilde{2}$, $\tilde{2}$, and $\{\tilde{2}/\tilde{2}\}$ of any property $f(t)$ (as indicated by Eqs. (1.1)–(1.3)), the pertinent outputs are *directly referred to their genetic origin*. In this way we also assert that they will keep "memory" of their genetic origin because, besides their specific quantities, they will result as being structured in Ordinal Terms, according to the indication of the specific exponent adopted. The latter is correspondently termed as *Ordinality*, because it precisely expresses how each part of the output is genetically Ordered to the Whole (and, consequently, but only as a posterior concept, also related to all the others).

A simple example of an Ordinal Co-Production and, at the same time of a non-conservative Algebra, is represented by the *genesis of two brothers*. These are properly defined as being "brothers" not because of their direct relationships between them, that is: because they respect each other or they love each (but they might also hate each other). They are "brothers", *in essence*, because generated by the same father (or the same mother). That is, because of their "irreducible" Relationship with the Generative cause of their being born.

Their genetic Relationship represents in fact something which is *unique*, *specific* and *irreducible*, so that they cannot simply be accounted for as "two" (1 + 1) simple persons (as if they were extraneous to each other), but as *one sole entity* (that is, as a Whole), albeit their clear reciprocal distinction. This is the reason for the term "*brothers*". A term which precisely accounts for that "extra", represented by their specific Relationship with the same genetic Principle.

On the other hand even in any Civil Law the relationship between two brothers is considered as of the second order. Precisely because the first order relationship is that which relates each one of them to the same father (or mother).

At this point the output Ordinal "Relationships" corresponding to Eqs. (1.1), (1.2), and (1.3) ("binary", "duet", and "binary-duet" Relationships, respectively) can simply be obtained on the basis of definition (A1), by considering an exponential Ordinal Relationship, in its most general form $e^{\tilde{\alpha}(t)}$.

It is well known, in fact, that any function $f(t)$ can always be written as

$$f(t) = e^{\ln f(t)} = e^{\alpha(t)}. \tag{A4}$$

In such a case we have

$$\left(\frac{\tilde{d}}{\tilde{d}t}\right)^{Ord} e^{\tilde{\alpha}(t)} = [\mathring{\alpha}(t)]^{Ord} \cdot e^{\tilde{\alpha}(t)} \tag{A5}$$

where $\mathring{\alpha}(t)$ represents the first order *incipient* derivative of $\tilde{\alpha}(t)$. The symbol of a little circle adopted to denote the incipient derivative was evidently chosen in analogy to classical Newton's "dot" notation, which usually indicates a first-order derivative.

The different symbology is here justified by the fact that the former should now remind us the conceptual difference between the incipient derivative and the traditional one.

In fact, even if $\mathring{\alpha}(t)$ and $\tilde{\alpha}(t)$ coincide (but only) from a pure cardinal point of view, they are, on the contrary, radically different from a Generative point of view. The former in fact represents the specific exit of a Generative Process, whereas the latter is always understood as the result of a necessary process (thought of as being a "mechanism" or a set of "mechanisms"). On the other hand such a quantitative coincidence is strictly valid only for *Ord* = 1.

The definition of "incipient" derivative, in fact, is valid for *any* fractional number *Ord*. This means that we may adopt the same definition to model *any* complex System, by simply considering "incipient" derivatives characterized by those rational numbers $(\tilde{m}/\tilde{n})$ which result as being more appropriate to each specific System analyzed (see, for instance, Eq. (2) in the text).

In such a case, the right hand side of Eq. (A5) reveals an extremely important property: a sort of "*persistence of form*", which is particularly marked when the derivative is of fractional order $(\tilde{m}/\tilde{n})$. This precisely represents that fundamental property which leads to the formulation of *Harmony Conditions* (8) and (9).

The latter in fact represent a simple "adherent" consequence of any Generative Process, although characterized by its specific generation modalities. In other words, any "Generating Process" (modelled by the left hand side of Eq. (A5)) gives origin to an output which corresponds to a multiple structure Relationships (see Eq. (3)), characterized by the Ordinality expressed by the right hand side of the same Eq. (A5). All these Ordinal Relationships are then similar to harmonic evolutions always in "resonance" (as in a "musical chord") with the their unique and irreducible Genetic Process and, at the same time, with each other. They reach their *maximum* harmony only when they, *independently from the specific phenomenological field analyzed*, are related to each other in form (23). That is,

in terms of the corresponding Ordinal Roots of Unity (see Eq. (25) and (26)).

A.3. A simple example of Emerging Solution

Any System modelled according to the M.O.P. is always described by an Emerging Solution whose formal structure is represented by Eq. (10), where all the elements of the Ordinal Matrix emerge from Eqs. (8) and (9). These elements are always of Ordinality $\{\{\tilde{2}/\tilde{2}\} \uparrow \{\tilde{2} \uparrow\}\}$ because they also satisfy *specularity* Relationships (7)).

Such an Emerging Solution *can never be expressed* in terms of traditional "functions", because, if we try to do that, we end up by systematically "filtering" the three basic assumptions of the M.O.P. (*Generative Causality*, *Adherent Logic*, *Ordinal Relationships*) and, consequently, by "reducing" the description to the traditional assumptions of *efficient causality*, *necessary logic*, *functional relationships*. In such a way, we not only profoundly alter the previous conceptual description of the System but, in addition, we get a final mere cardinal description of the same which is characterized by some (often well-marked) quantitative "drifts".

A very simple example is represented by any "Two-body problem" (think of, for instance, the Sun and Mercury).

The sequence of the various passages that "*reduce*" the corresponding Emerging Solution to a traditional functional solution is explicitly given in (Giannantoni, 2008b, ch. 22). In such a case the exponential Ordinal Matrix is progressively reduced to a linear combination of eight exponentials, which represent the cardinal reciprocal motion of the two bodies.

The result so obtained, however, can only be considered as being a simple better approximation with respect to the traditional solutions to Mercury's Precessions. This is because: (i) we have "extracted" only two bodies from the Solar System; (ii) consequently, the (mere quantitative) solution presents a much more marked "drift" when compared to the Emerging Solution pertaining to the Solar System considered as a Whole; (iii) in addition, we cannot clearly recognize the associated Emerging Exits (and the corresponding Ordinal "Drifts" with respect to them), which might indicate that the Solar System, in turn, is Ordinally related to the entire Universe, now more properly understood as a Whole.

This is precisely because we have reduced the intrinsic Quality of the Process to a mere "functional" aprioristic perspective, which is the primary cause of many insolvable problems (think of, for example, the "Three Body Problem" or the intrinsic instability of Systems designed as they were mere "mechanisms").

A.4. Some advantages of the model proposed compared to other models based on Neo-classic Economics

The "Three Good, Two Factor Problem" is a famous problem in Neo-Classical Economics (NCE).

It concerns the search for an *equilibrium* condition, between three different goods, in a competitive market and in the presence of two sole productive factors (Capital and Labour), but, at the same time, the equilibrium should also represent an *optimum* condition according to Pareto optimality.

The problem, which is intrinsically unsolvable in explicit terms, is generally considered as a "hard" problem in Computer Science, whereas, for specific (and very simple) conditions, it can even become an NP-Complete Problem.

This substantially depends on the fact that the fundamental Principles of NCE, such as Walras General Equilibrium, Pareto Optimality, etc., are the result of a direct transposition to economic activities of the Principles of Classical Mechanics and, even more, of Classical Thermodynamics.

The Utility-Expenditure Conservation Principle, for example, corresponds to Energy Conservation Principle, which (as previously shown in the text) does not hold when reconsidered in light of the M.O.P.

This also means that neither does Walras General Equilibrium represent a "stable" equilibrium condition nor does Pareto optimality represent a "maximum" condition, precisely because the latter presupposes the former.

This should clearly indicate that Economic Processes cannot be modelled as being mere "mechanisms" (as usually modelled by NCE).

The "Three Good, *Three* Factor Problem" is nothing but the reformulation of the former with the simple addition of a third productive factor, represented by Natural Resources. Correspondingly, the "*N* Good, Three Factor Problem" represents the widest generalization form of the latter, because extended to any number (*N*) of goods.

All these problems, when reformulated in terms of the M.O.P. always have an explicit solution.

The solution to the "*N* Good, Three Factor Problem", in fact, includes the solution to the "Three Good, *Three* Factor Problem" and, as a particular case, that to the "Three Good, *Two* Factor Problem" too.

The relevance of such explicit solutions can simply be understood by considering the possibility of strategic decisions in Economics (especially in International Trade, but not only). The explicit solutions to such problems, in fact, enable a Decision Maker (at any level) to progressively realize, in actual fact, the Maximum Ordinality conditions which, at the same time, correspond to the maximum intrinsic stability of the Economic System analyzed, because the latter are a simple Ordinal reflex of the associated Harmony Relationships.

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www.ordinality.org, 2012. Author's website that presents a general framework about the M.O.P., from the Mathematical Formulation of the Maximum Em-Power Principle up to the Mathematical Formulation of the M.O.P., together with the various Ostensive Examples mentioned in this paper.